6 (6)

Probability and Probability Distributions

Binomial Distributions

The binomial distribution is used when:

- There is a ***fixed number of trials*** (***n***)**.**
- Each trial has ***only two outcomes***: *Success* or *Failure* .
- The ***probability of success*** (***p***) is the same for each trial.
- Trials are ***independent.***

Binomial Distributions

combination of *r* success from *n* trials

Number of success **Number of failures**

$$P(X = r) = {}_nC_r\, p^r (1 - p)^{(n - r)},$$

Probability of success

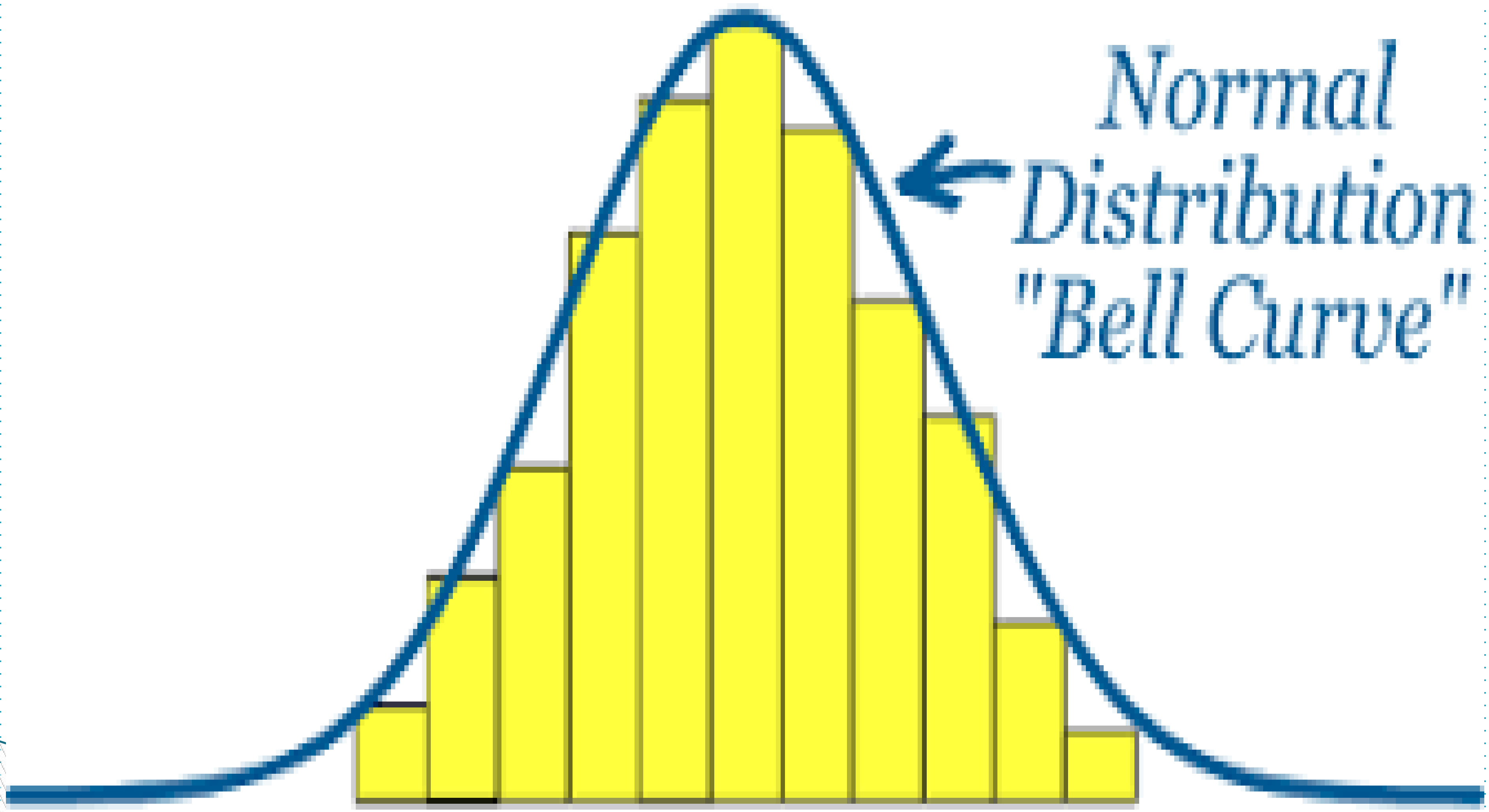
Random variable X **probability of failure**

Normal Distributions

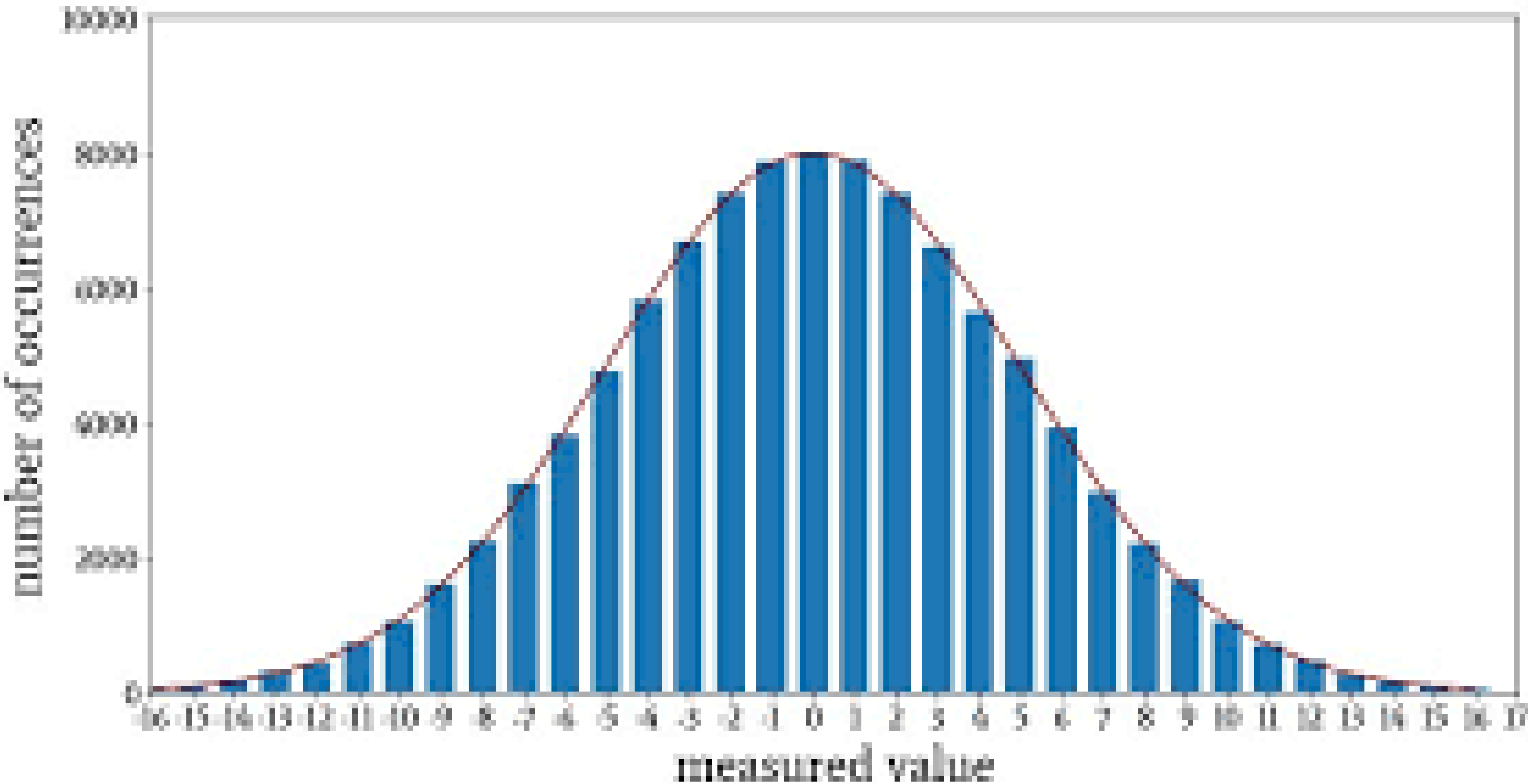
6.5 Normal Distributions

- It is one of the probability distributions a ***continuous random variable*** can possess.
- A large number of phenomena in the real world are normally distributed either exactly or approximately.

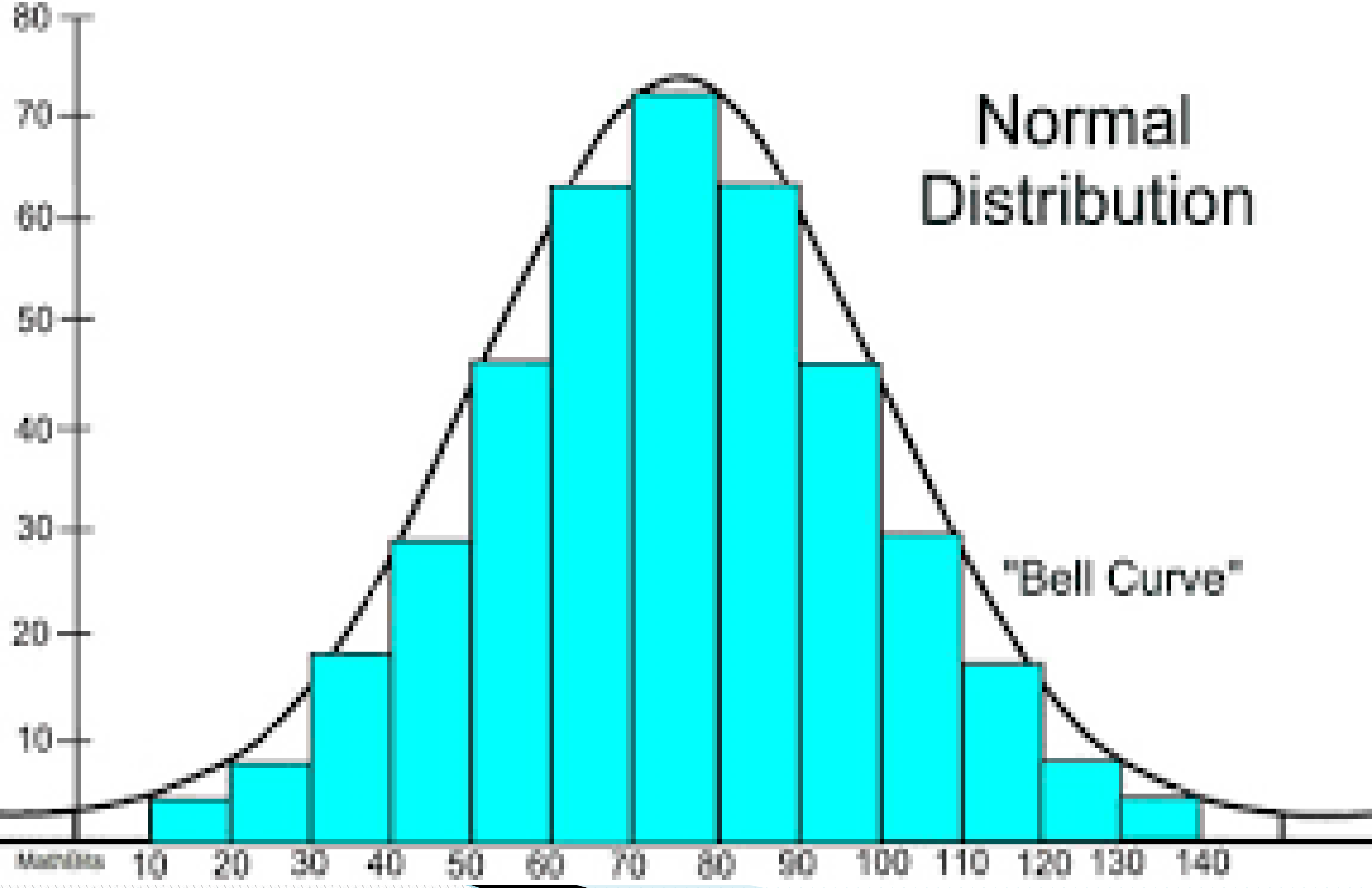
6.5 Normal Distributions



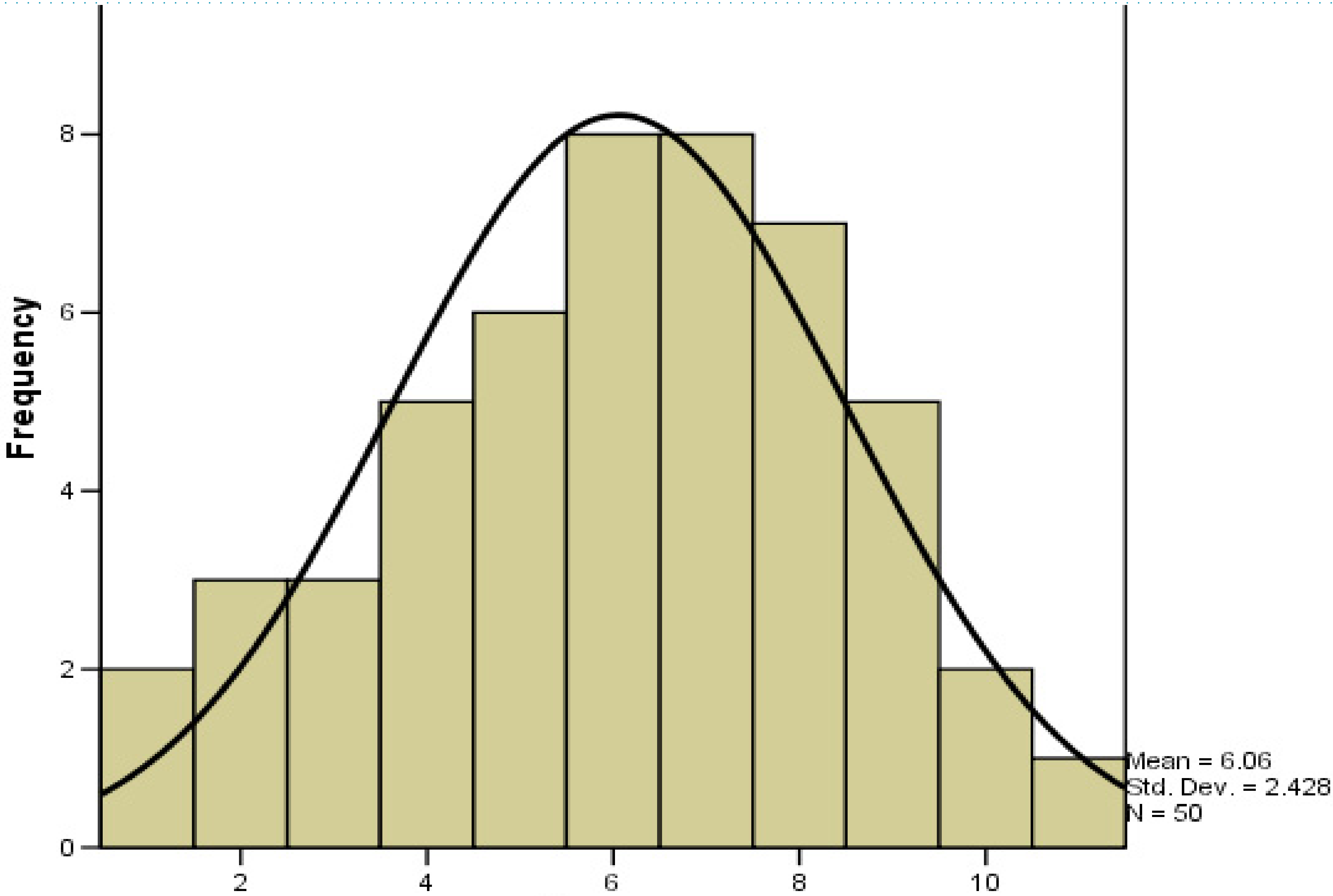
6.5 Normal Distributions



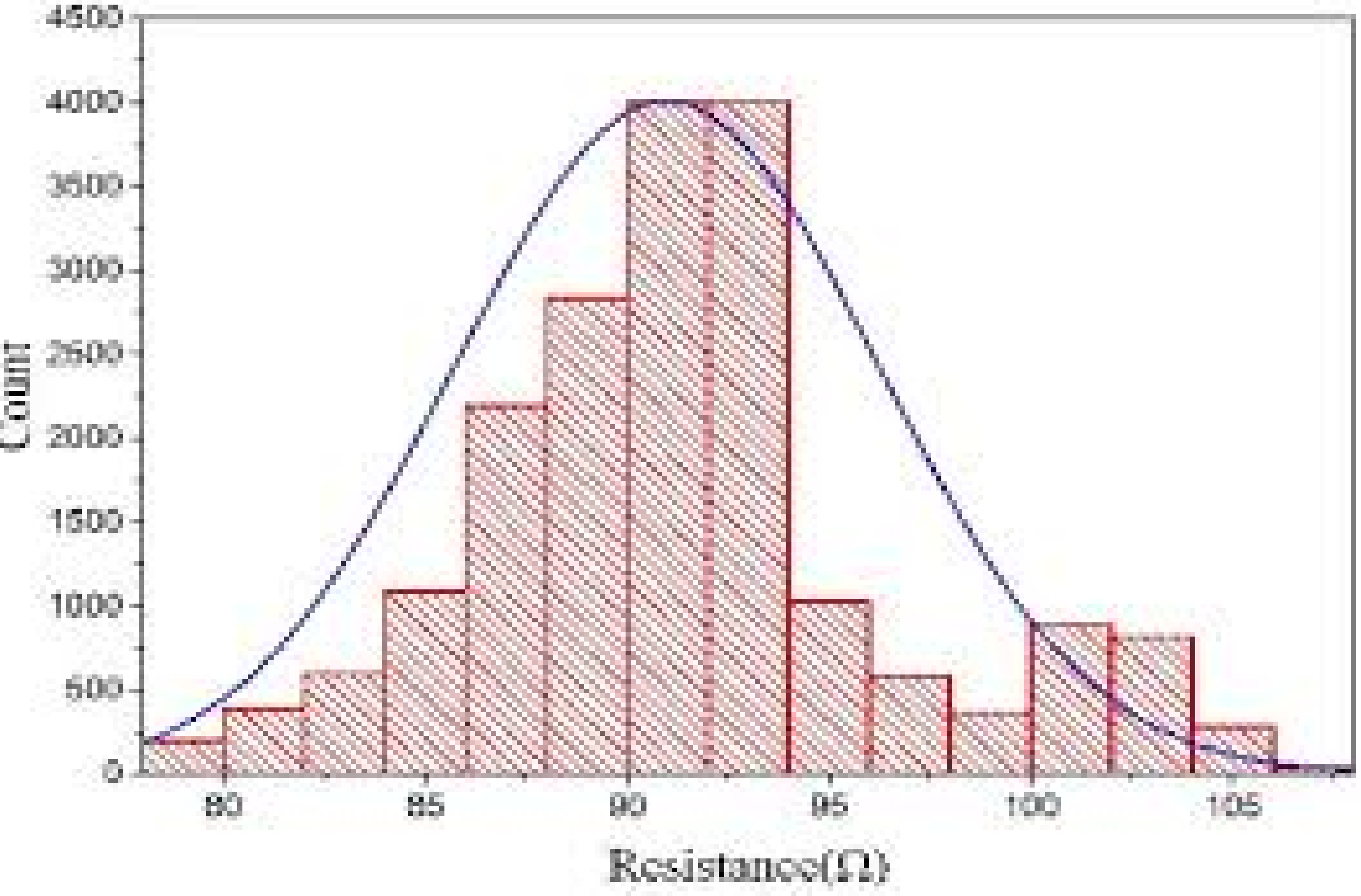
6.5 Normal Distributions



6.5 Normal Distributions



6.5 Normal Distributions

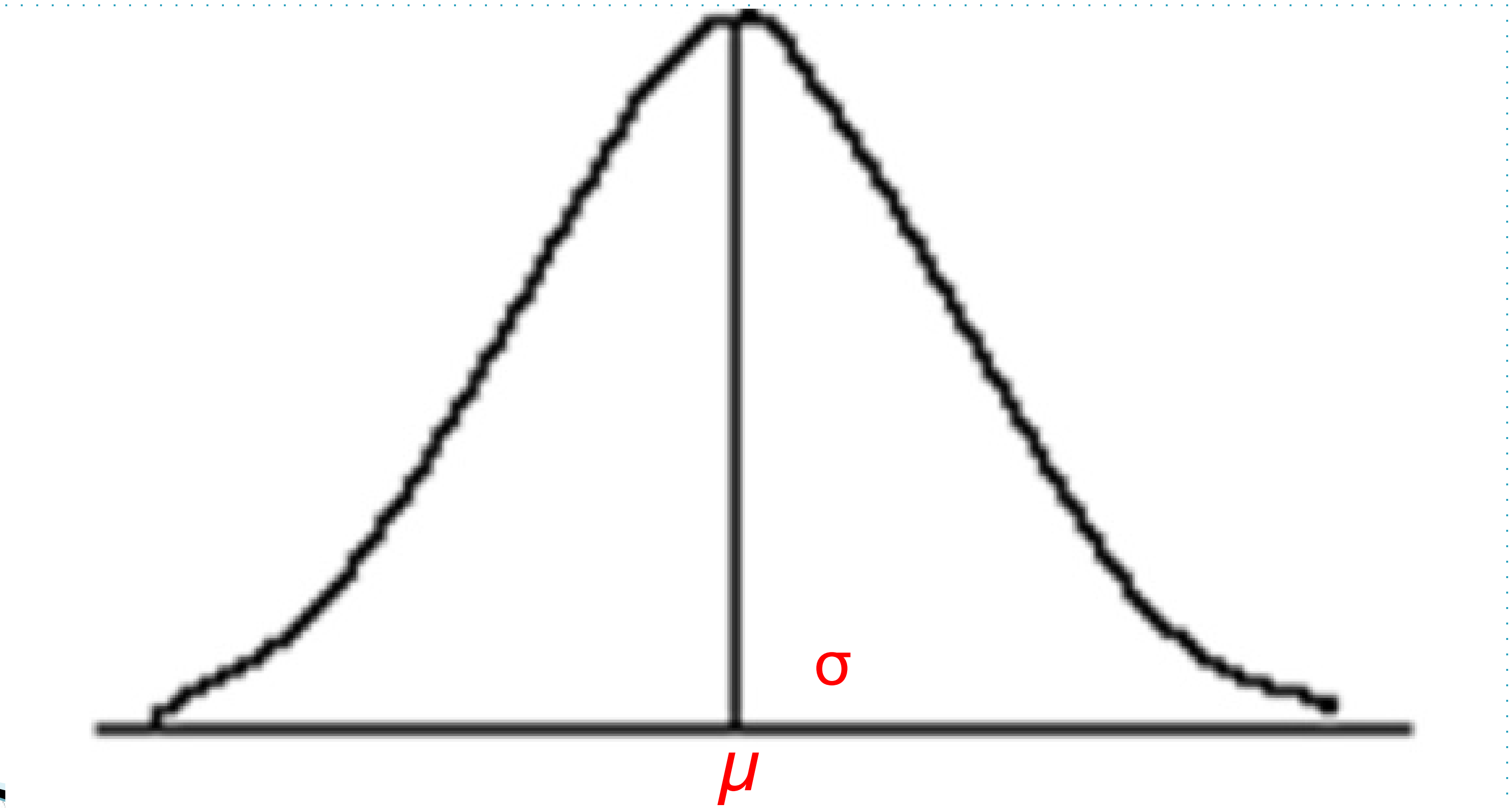


6.5 Normal Distributions (Contd. . . .)

- The normal probability distribution or the normal curve is ***a bell shaped (symmetric) curve.*** Its mean is denoted by μ and its standard deviation is denoted by σ.

6.5 Normal Distributions (Contd. . . .)

Normal distribution with mean μ and standard deviation σ.



6.5 Normal Distributions (Contd. . . .)

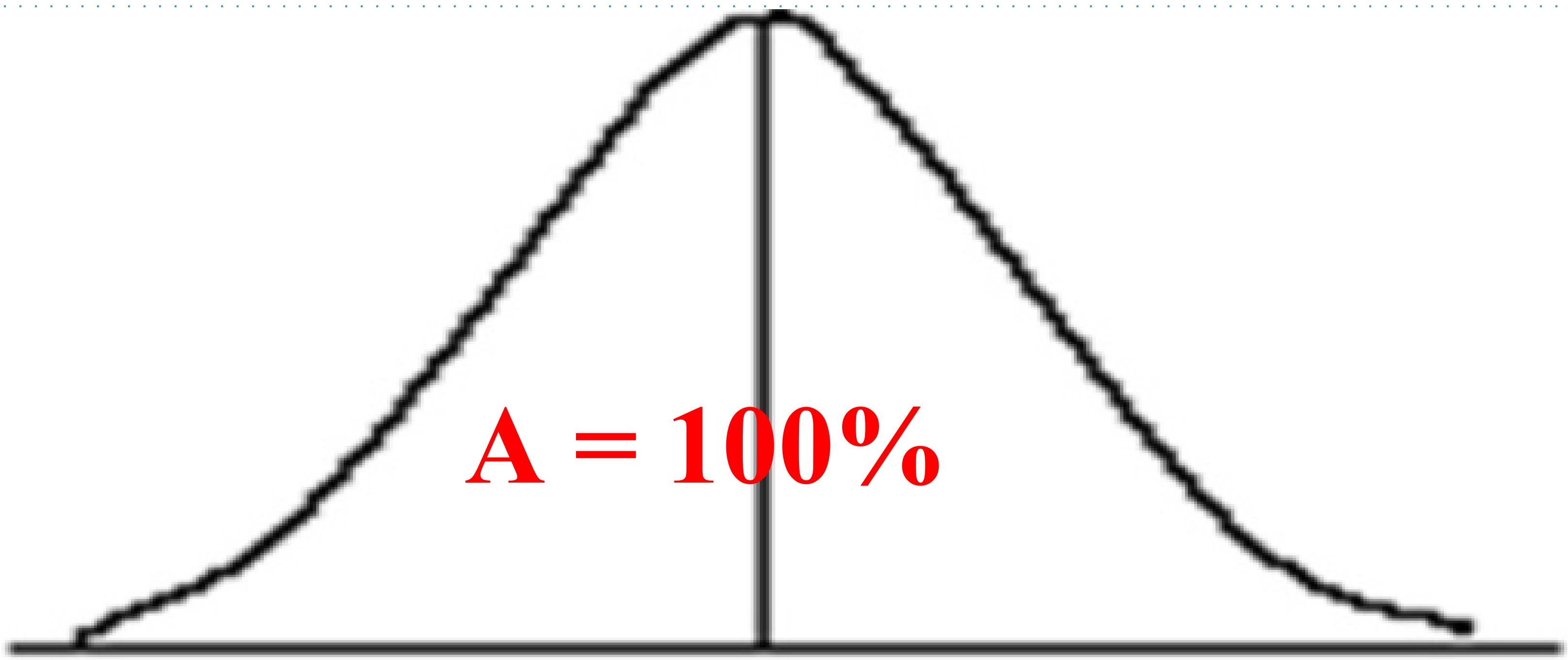
- ***Characteristics of a Normal Distribution:***

1. The total area under a normal distribution curve is 1 or 100%.

2. A normal distribution curve is ***symmetric about the mean*** (half of the total area under a normal distribution lies on the *left side* of the mean, and half lies on the *right side* of the mean).

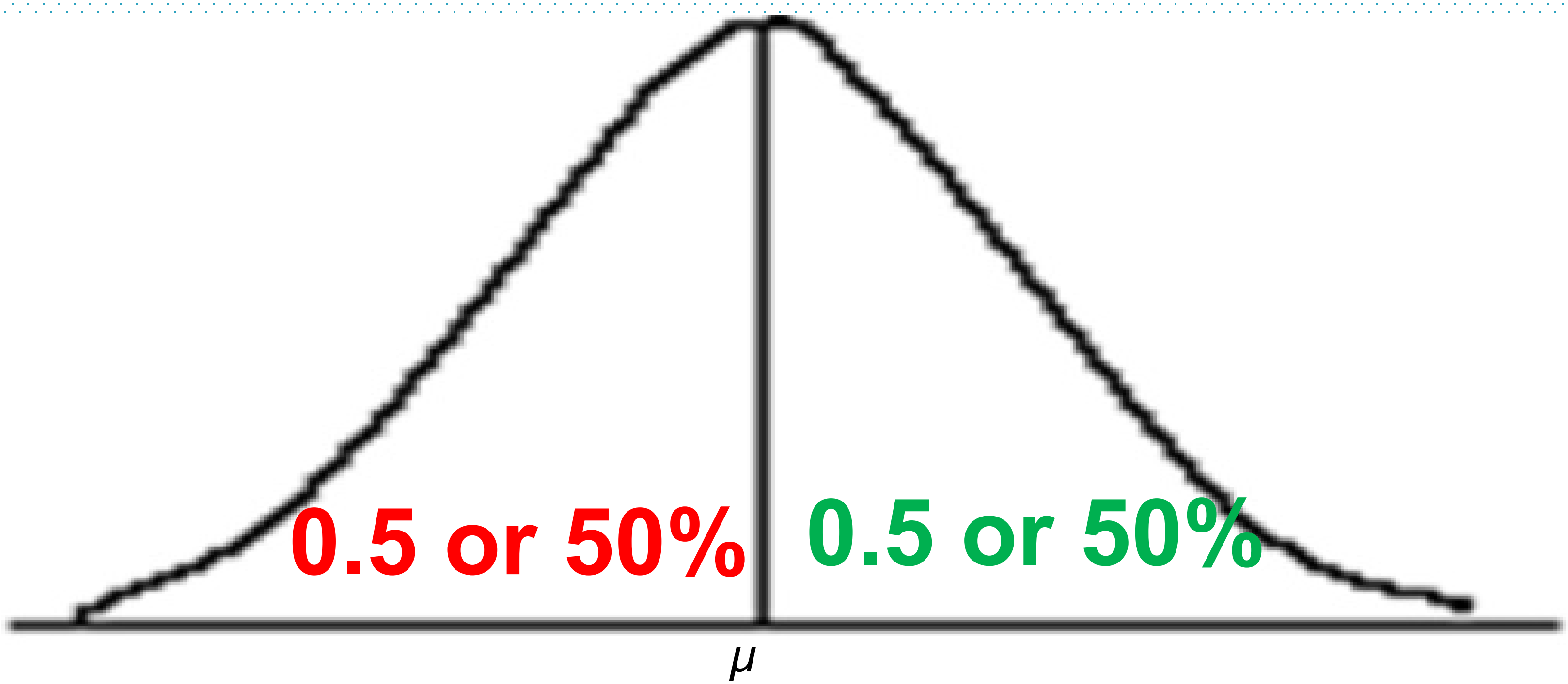
6.5 Normal Distributions (Contd. . . .)

The area under a normal curve is 1.0 or 100%



6.5 Normal Distributions (Contd. . . .)

A normal curve is symmetric about the mean



6.5 Normal Distributions (Contd. . . .)

3. The tails of a normal distribution curve extends indefinitely in both directions without touching or crossing the horizontal axis.

6.5 Normal Distributions (Contd. . . .)

- Although a normal distribution curve never meets the horizontal axis, beyond the points represented by $\mu - 3\sigma$ and $\mu + 3\sigma$ it becomes so close to this axis that the area under the curve beyond these points in both directions can be taken as virtually zero.

6.9

The Standard Normal Distributions

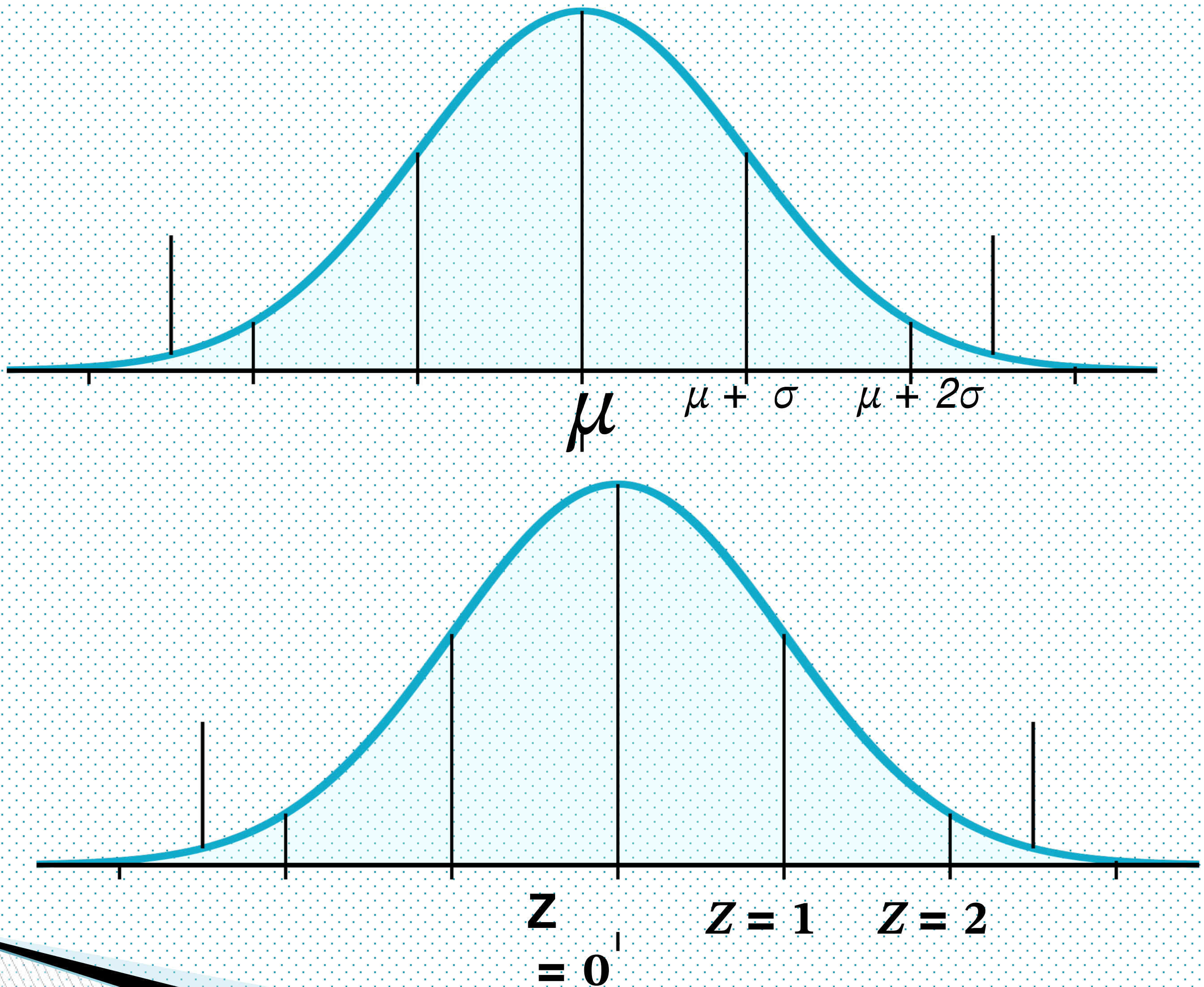
6.9 The Standard Normal Distributions

- The standard normal distribution is a *special case of the normal distribution.*
- For the standard normal distribution, the value of the mean is equal to zero, and the value of the standard deviation is equal to 1.

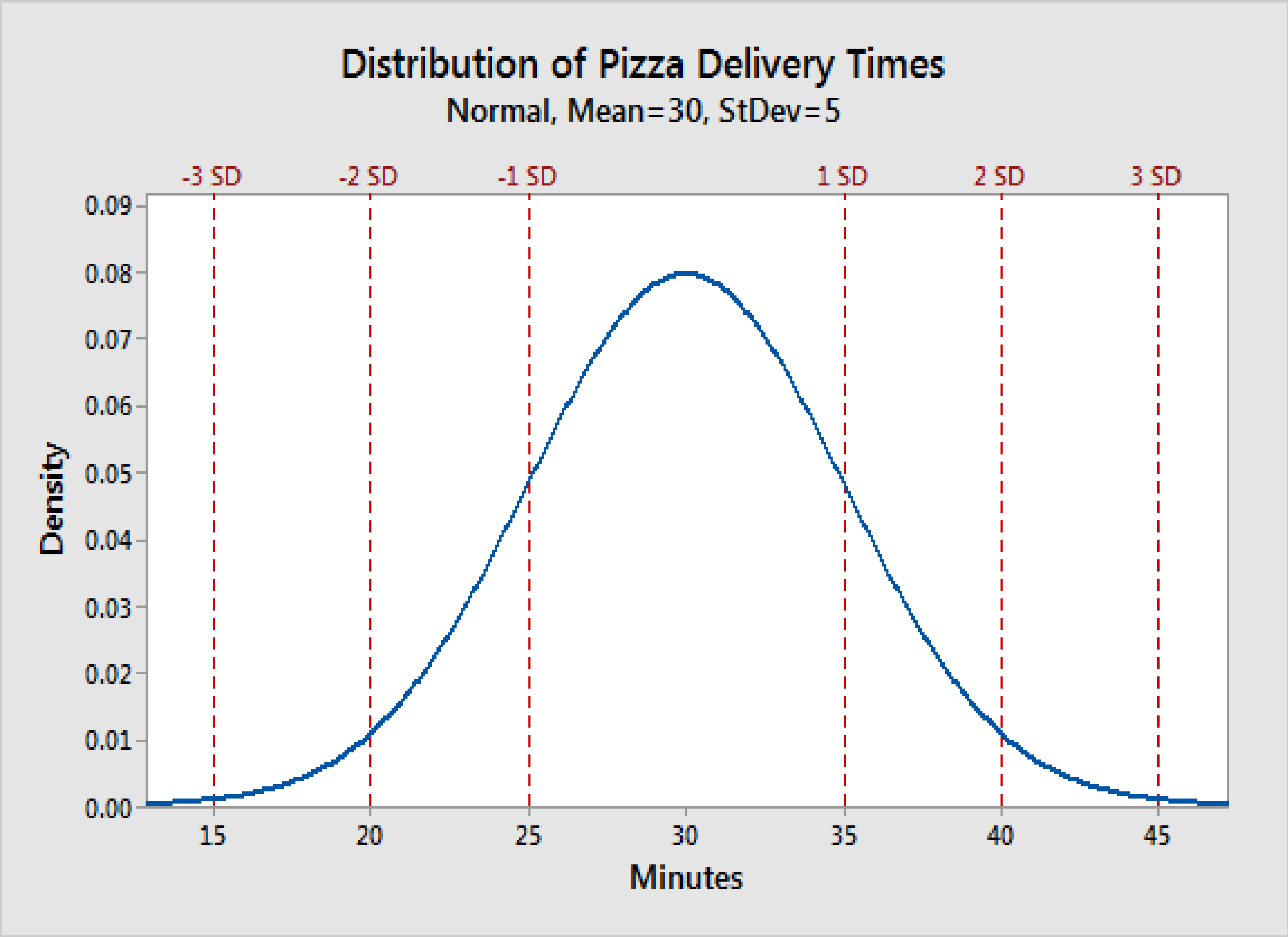
6.9 The Standard Normal Distributions (Contd. . . .)

- The standard normal distribution table, lists the area under the standard normal curve between $z = 0$ and the values of z from 0.00 to 3.09.
- Note always that although the value of z on the left side of the mean are negative, the area under the curve is always positive.

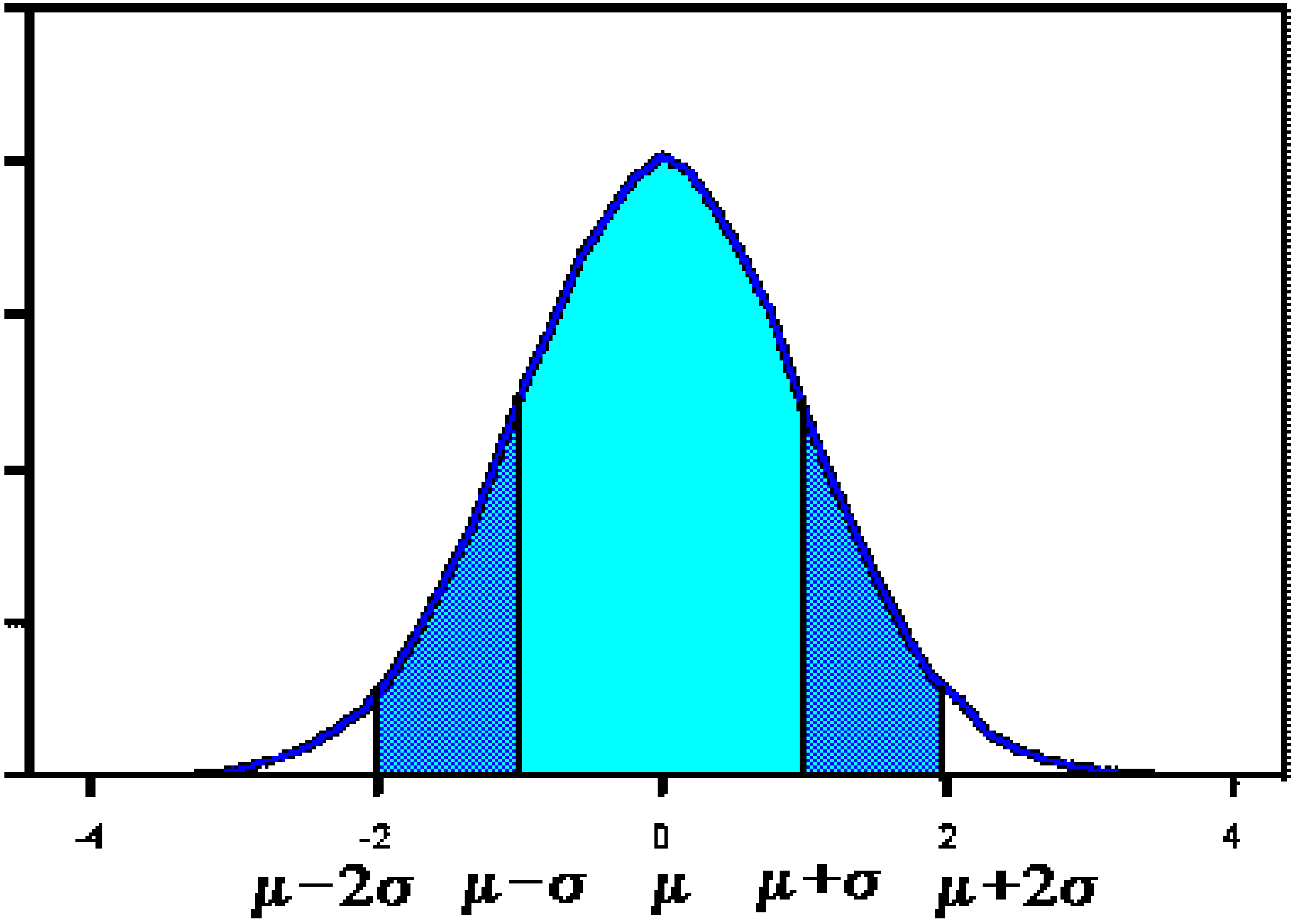
Application of Normal distribution



Application of Normal distribution



Application of Normal distribution



6.9 The Standard Normal Distributions (Contd. . . .)

Example 1:

i. Find the area under the standard normal curve from $z = 0$ to $z = 1.95$.

ii. Find the area under the standard normal curve from $z = -2.17$ to $z = 0$.

iii. Find the area under the standard normal curve to the right of $z = 2.32$.

6.6 The Standard Normal Distributions (Contd. . . .)

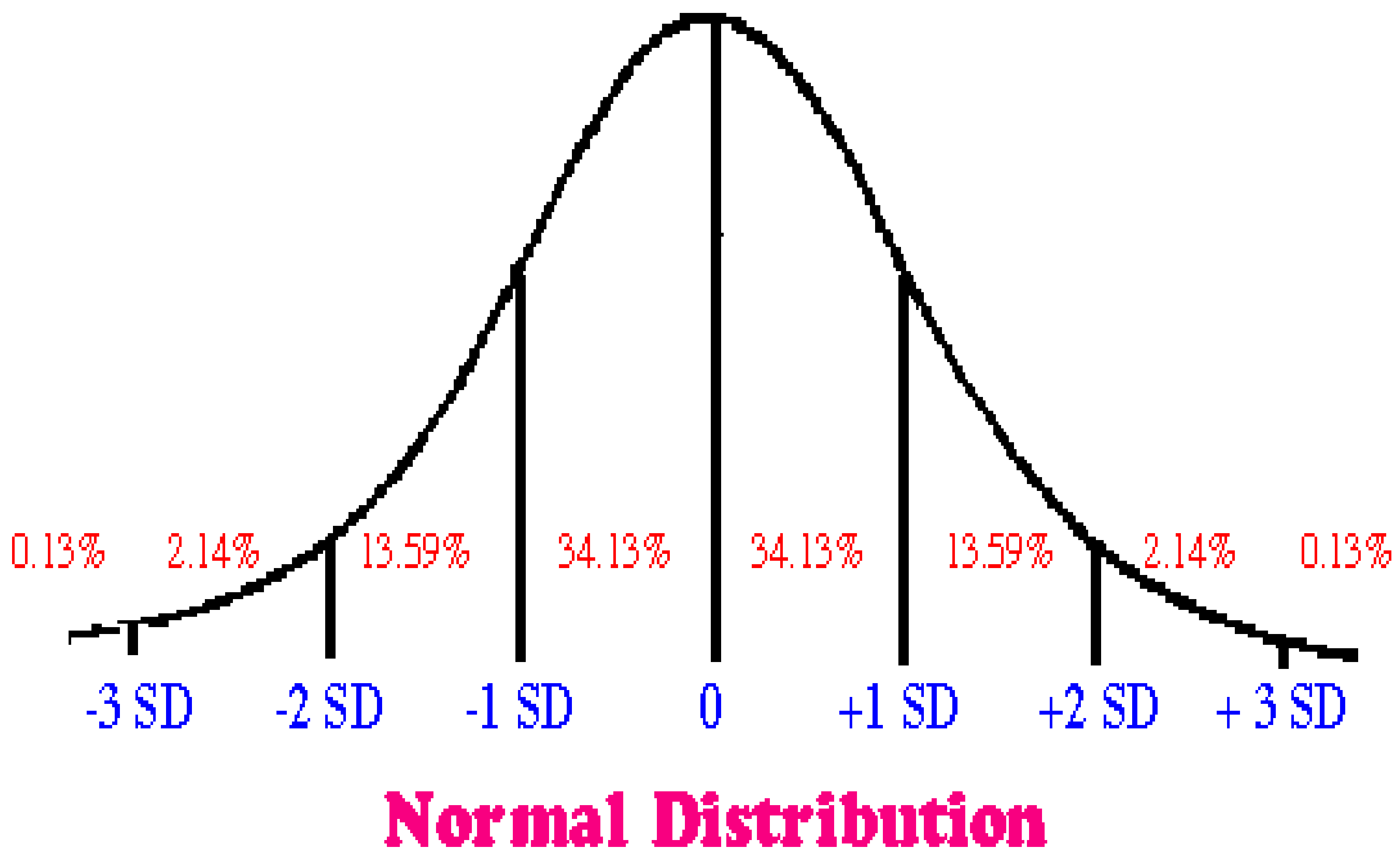
iv. Find the area under the standard normal curve to the left of $z = -1.54$.

Note that the area between $z = a$ and $z = b$ gives the probability that z lies in the interval a to b.

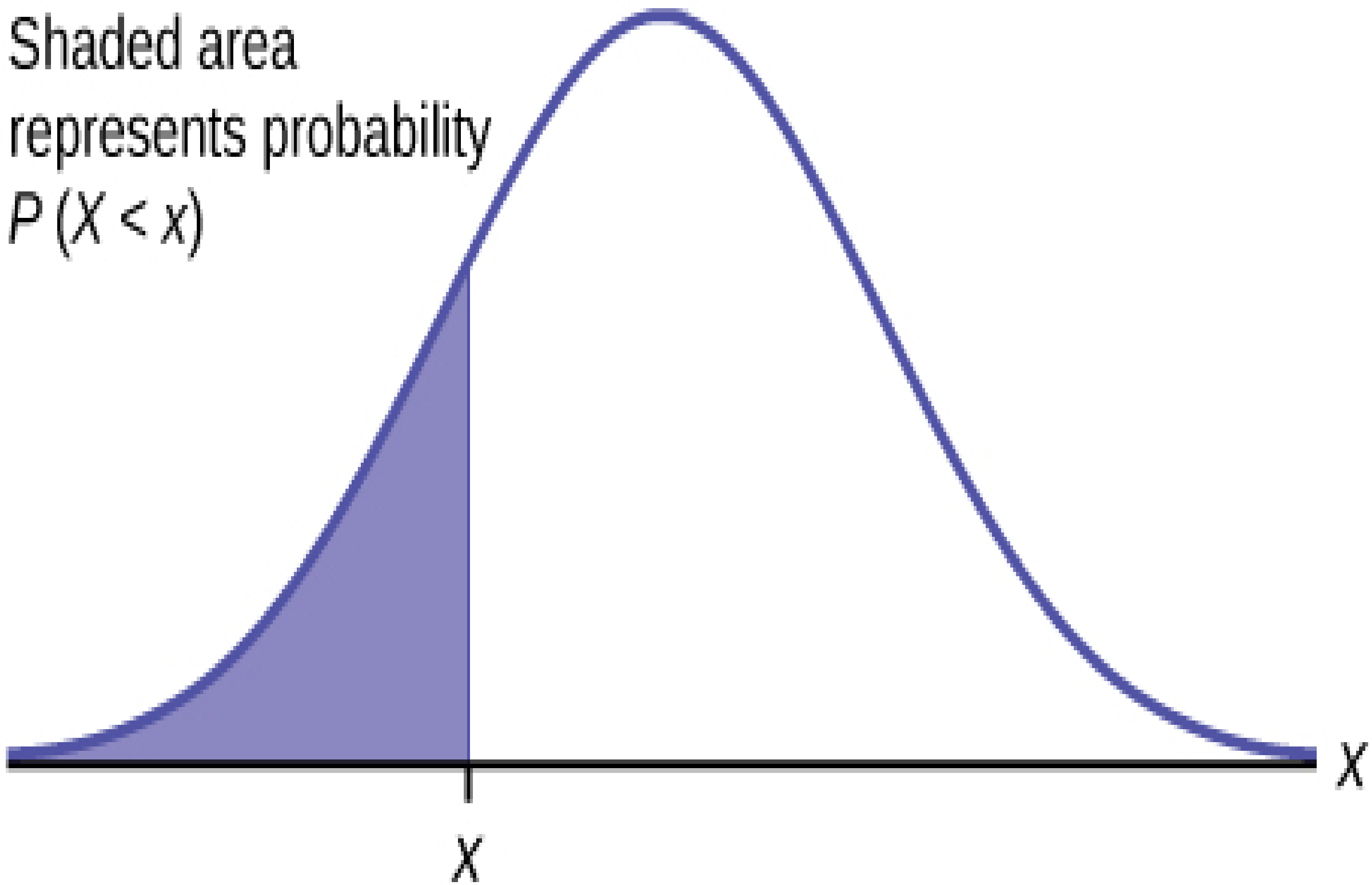
i.e. (Area from $z = a$ to $z = b$) $= p(a \leq z \leq b)$.

v. Find the probability $p(1.19 < Z < 2.12)$ for the standard normal curve.

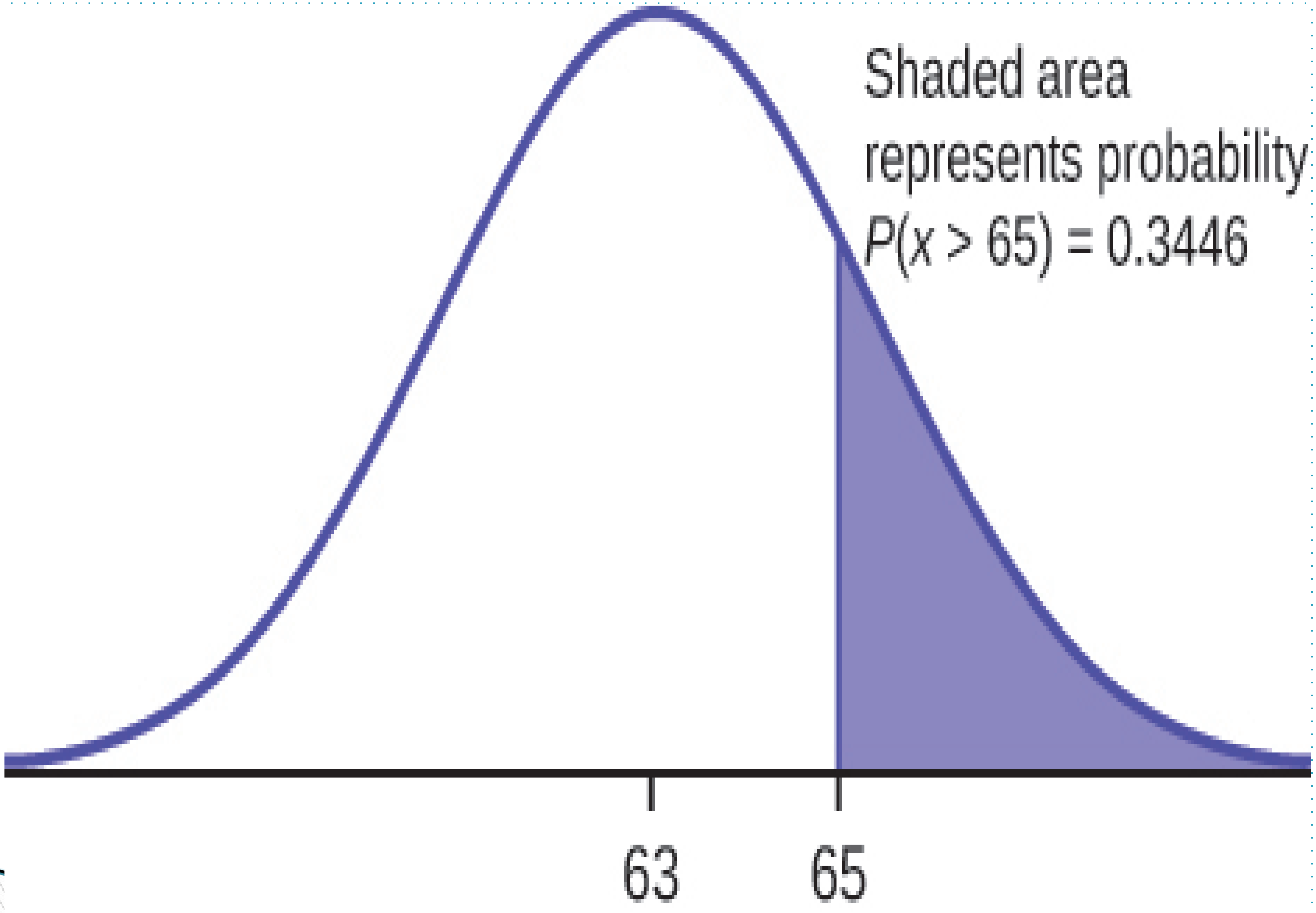
Application of Normal distribution



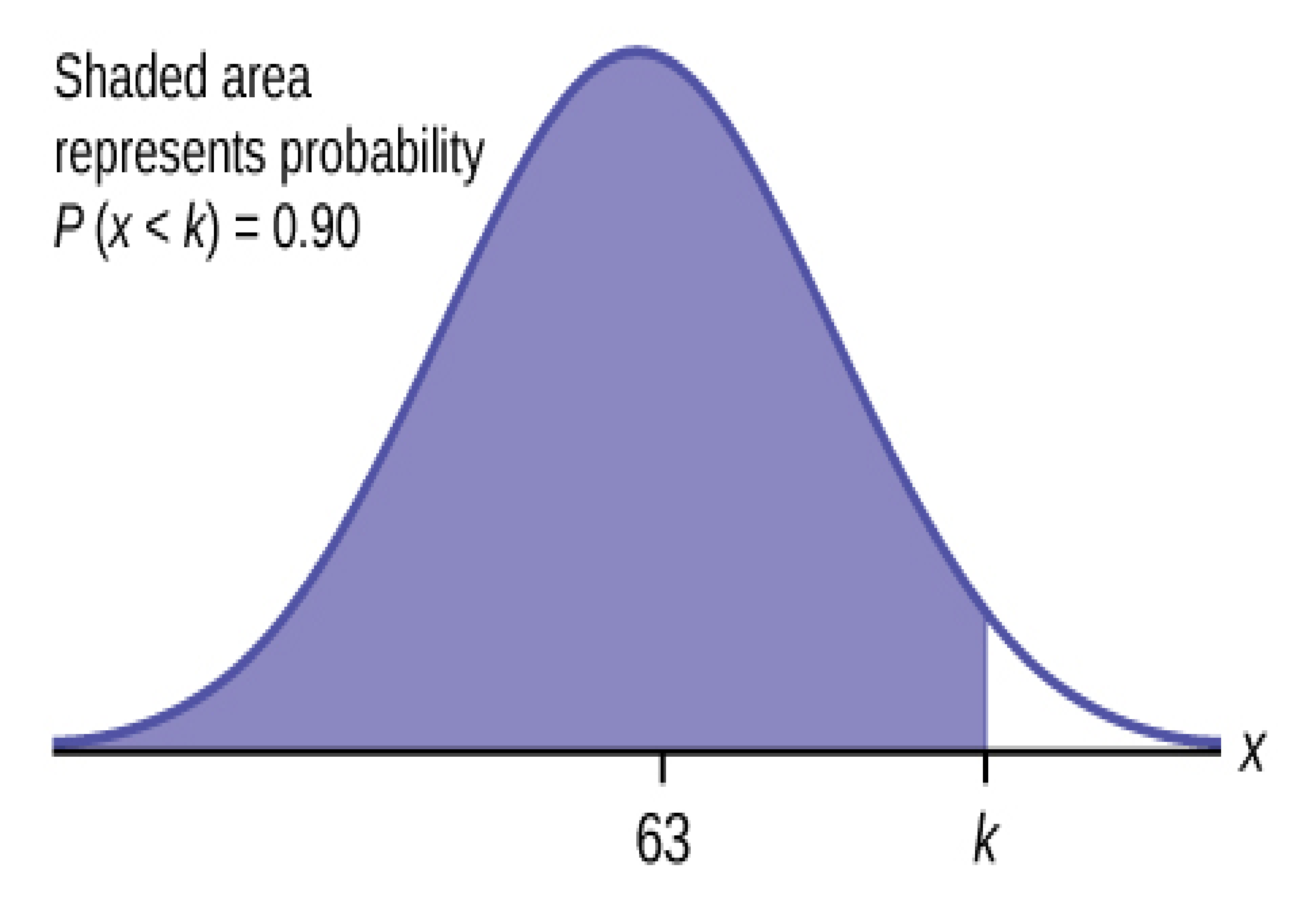
Application of Normal distribution



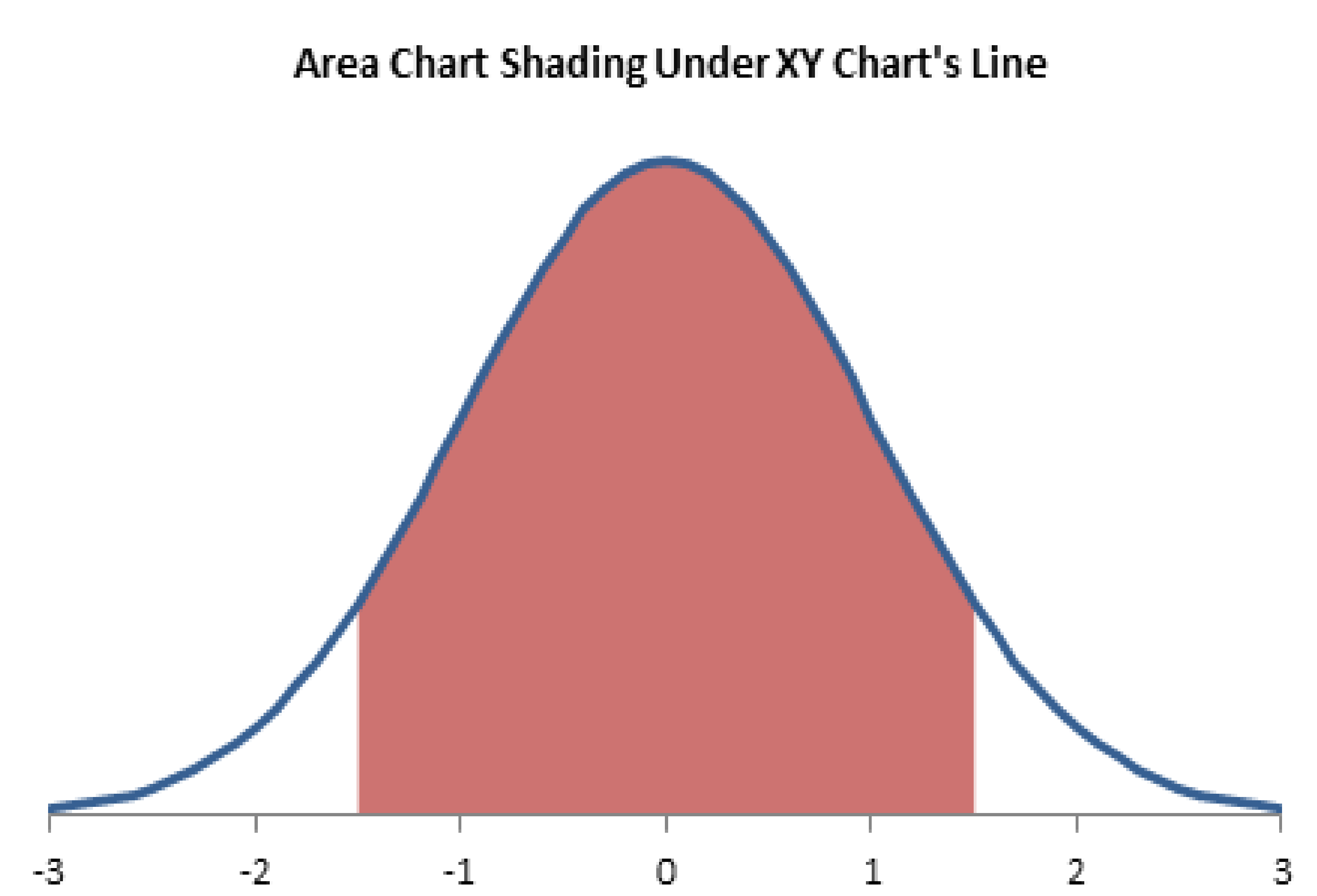
Application of Normal distribution



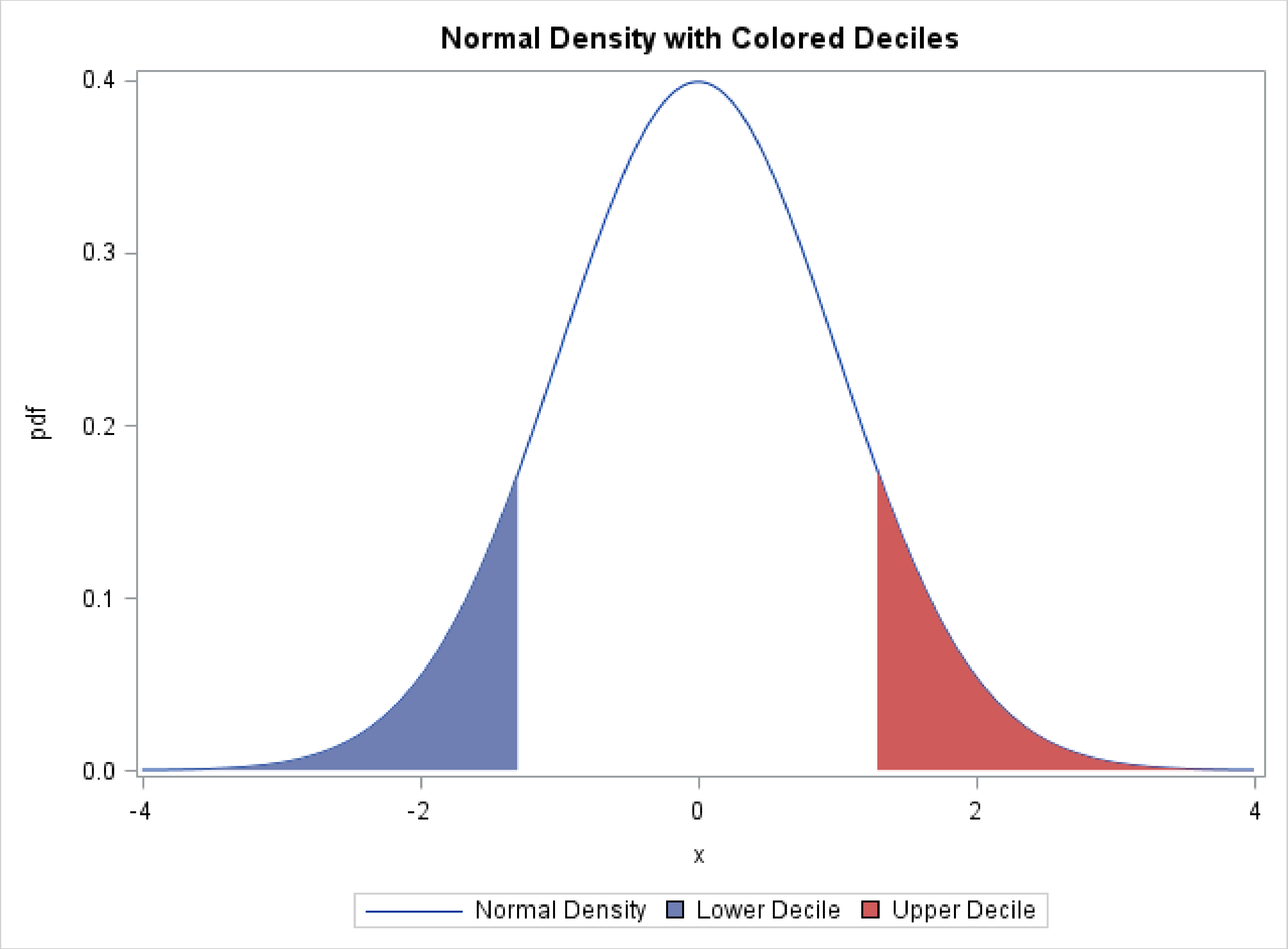
Application of Normal distribution



Application of Normal distribution



Application of Normal distribution



Empirical Rule:

For a bell-shaped distribution, approximately:

1. *Approximately* 68% of a distribution lie within one standard deviation (actually it is 68.26%)
2. *Approximately* 95% of a distribution lie within two standard deviation (actually it is 95.44%)
3. *Approximately* 99.7% of a distribution lie within three standard deviation. (actually it is 99.74%)

Empirical Rule (Contd. . . .)

Example 2:- The age distribution of a sample of 5000 individuals is a bell-shaped with a mean of 40 years and a standard deviation of 12 years. Determine the approximate percentage of people who are 16 to 64 years old.

Converting an x value to a z value

For a normal random variable x, a particular value of x can be converted to its corresponding Z value by using the formula: $Z = \dfrac{x - \mu}{\sigma}$,

where μ and σ are the mean and standard deviation of the normal distribution of x, respectively. Note that the Z value for the mean of a normal distribution is

Converting an x value to a z value (Contd. . . .)

Example3:- Let x be a continuous random variable that has a normal distribution with a mean of 50 and a standard deviation of 10. Convert the following x values in to z values.

a) $x = 55$

b) $x = 35$

Converting an x value to a z value (Contd. . . .)

Solution3:- ***a)*** $Z_{55} = \frac{x - \mu}{\sigma}$, where $x = 55$, $\mu = 50$ and $\sigma = 10$. Thus $Z_{55} = \frac{55 - 50}{10} = \frac{5}{10} = 0.5$

b) $Z_{35} = \frac{x - \mu}{\sigma}$, where $x = 35$, $\mu = 50$ and $\sigma = 10$. Thus $Z_{55} = \frac{35 - 50}{10} = \frac{-15}{10} = -1.5$

Converting an x value to a z value (Contd. . . .)

Example 4:- Let x be a normal random variable with a mean of 40 and standard deviation of 5. Find

a) $p(x > 55)$ b) $p(x < 49)$

Solution 4:- a) First we have to find the Z value of $x = 55$.

$$Z_{55} = \frac{x - \mu}{\sigma}$$, here $x = 55$, $\mu = 40$ and $\sigma = 5$.

Thus $$Z_{55} = \frac{55 - 40}{5} = \frac{15}{5} = 3$$

Converting an x value to a z value (Contd. . . .)

Now we read the Z value of $Z = 3$ from the Standard Normal Table which gives us 0.4987.

Thus $P > 55$ will be the area bounded by the region $Z > 3$ which can be found by subtracting 0.4987 from 0.5 which is the area for $Z > 3$.

$\therefore [P > 55] = 0.5 - 0.4987 = 0.0013$

Converting an x value to a z value (Contd. . . .)

Thus $P > 55$ will be the area bounded by the region $Z > 3$ which can be found by subtracting 0.4987 from 0.5 which is the area for $Z > 3$.

$\therefore [P > 55] = 0.5 – 0.4987 = 0.0013$

Converting an x value to a z value (Contd. . . .)

Solution 4:- b) First we have to find the Z value of $x = 49$.

$$Z_{49} = \frac{x - \mu}{\sigma}$$, here $x = 49$, $\mu = 40$ and $\sigma = 5$.

Thus $$Z_{49} = \frac{49 - 40}{5} = \frac{9}{5} = 1.8$$

Now we read the Z value of Z = 1.8 from the Standard Normal Table which gives us 0.4641.

Converting an x value to a z value (Contd. . . .)

But 0.4641 is the probability or the area bounded by $Z = 0$ and $Z = 1.8$. Thus $P < 49$ will be the area bounded by the region $Z = 0$ and $Z = 1.8$ and the whole region to the left of $Z = 0$.

Thus $[P < 49] = 0.4641 + 0.5 = 0.9644$.


Any Questions
Any Questions


Thank
you